

Detailed Generalized Multi-particle Correlation Spectral Analysis

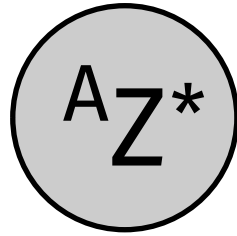
Bryan M. Harvey

ARUNA Seminar Series

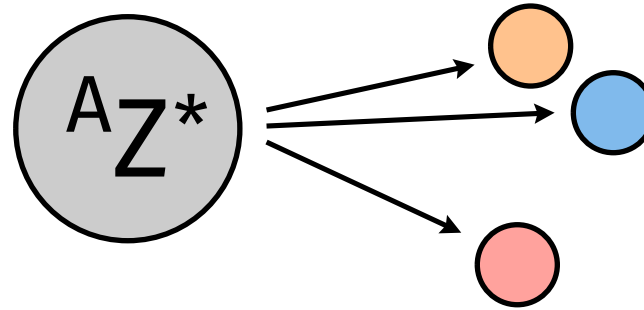
2026-09-21

Title

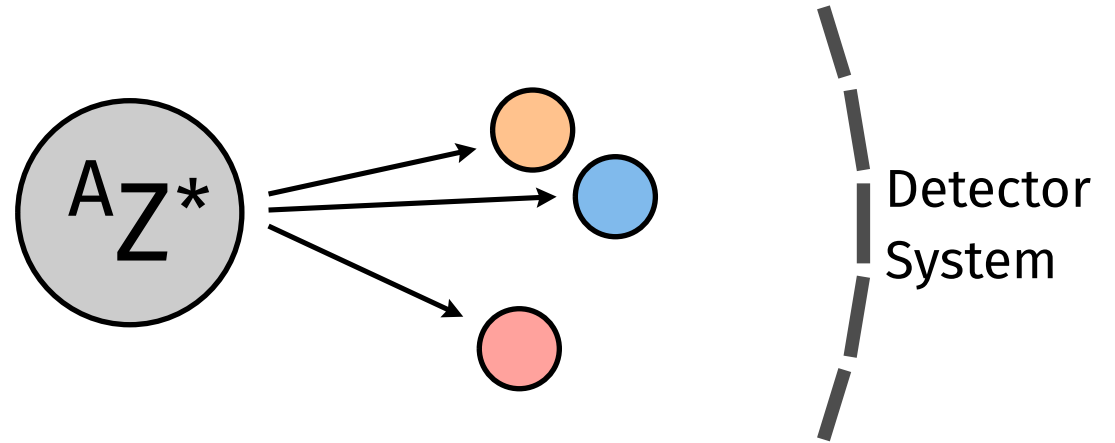
Multi-particle Correlations as a Tool



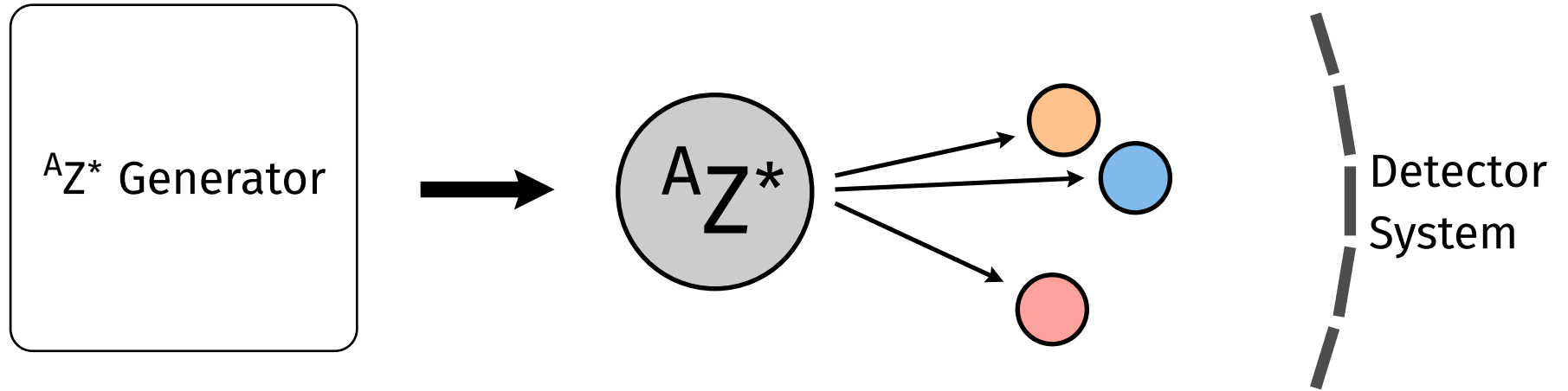
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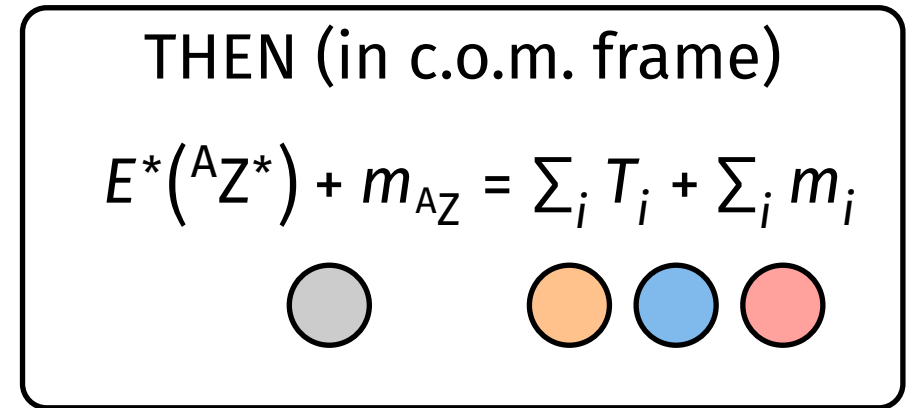
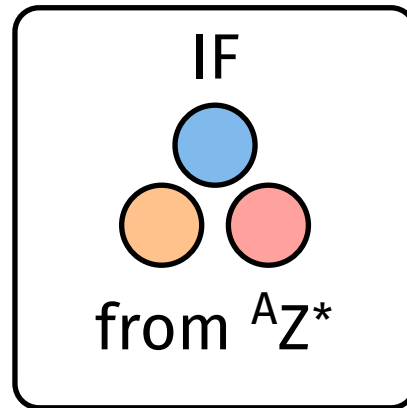
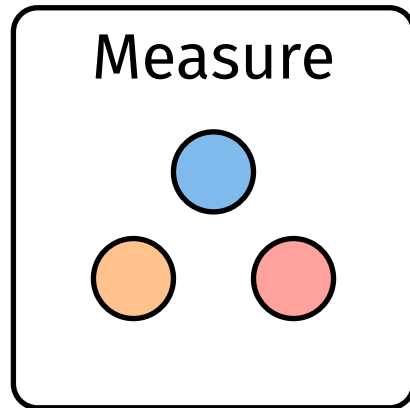
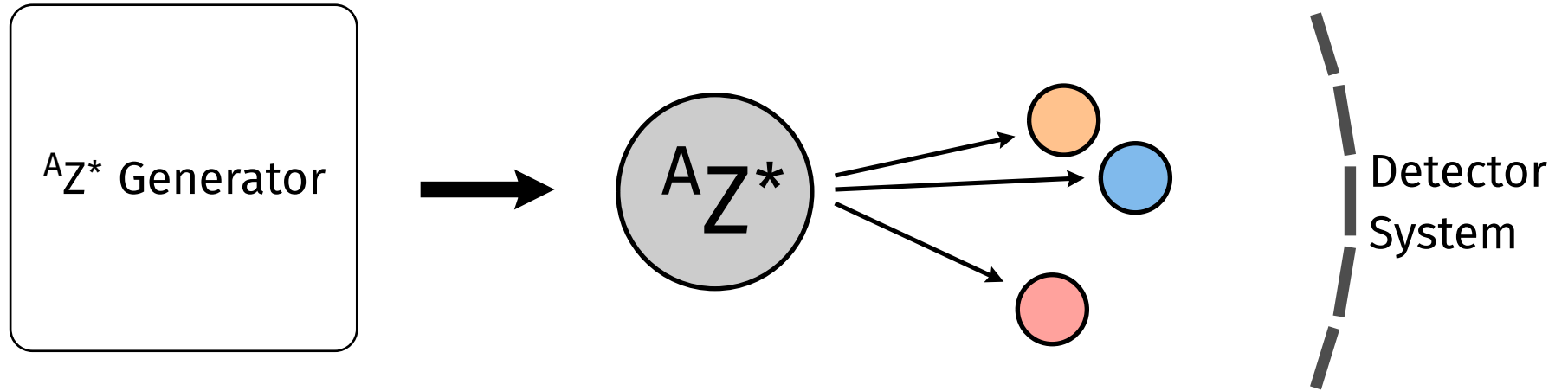
Multi-particle Correlations as a Tool



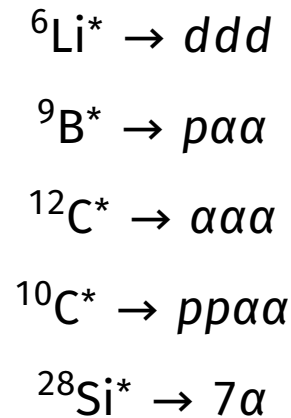
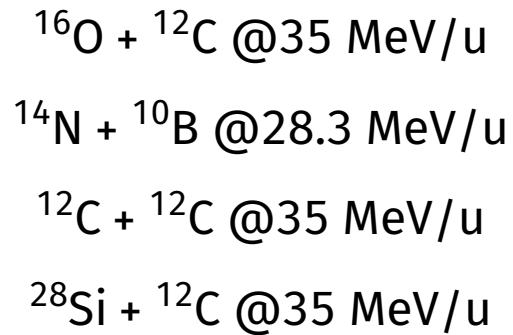
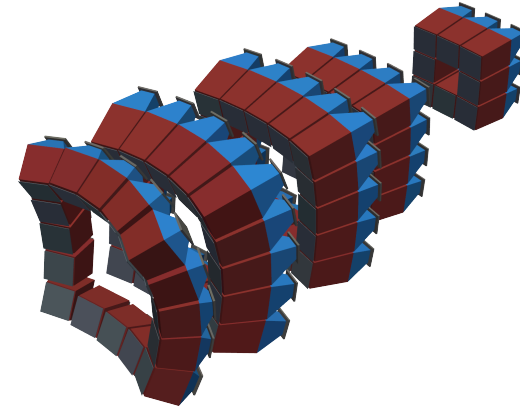
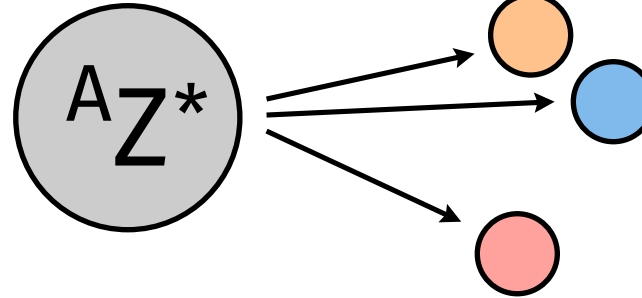
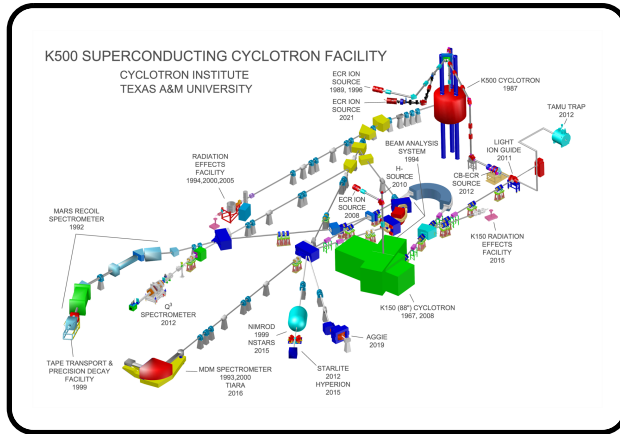
Multi-particle Correlations as a Tool



Multi-particle Correlations as a Tool

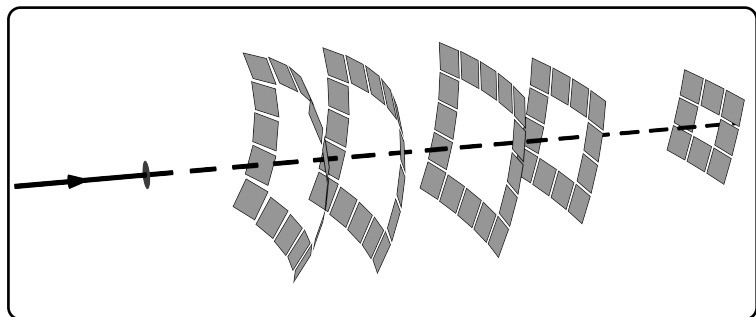
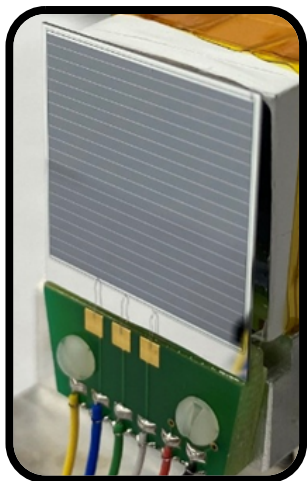
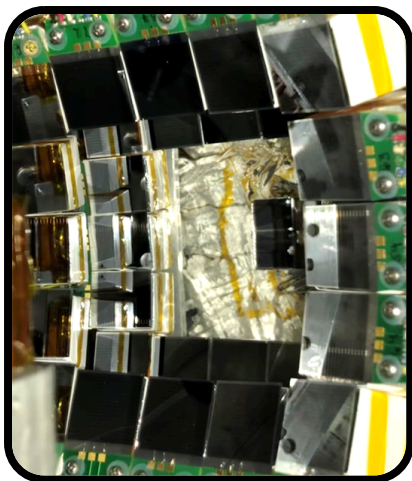


Multi-particle Correlations as a Tool



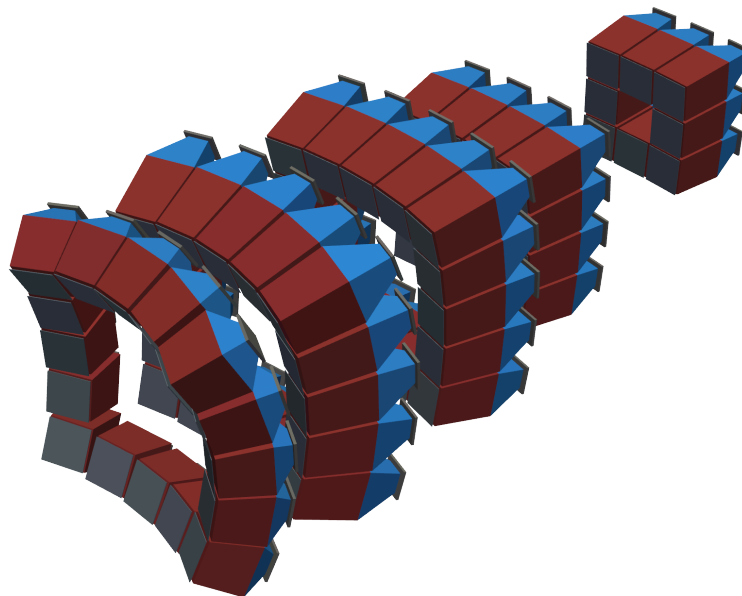
**Forward
Array
Using
Silicon
Technology**

Forward Array Using Silicon Technology (FAUST)



Forward Array Using Silicon Technology (FAUST)

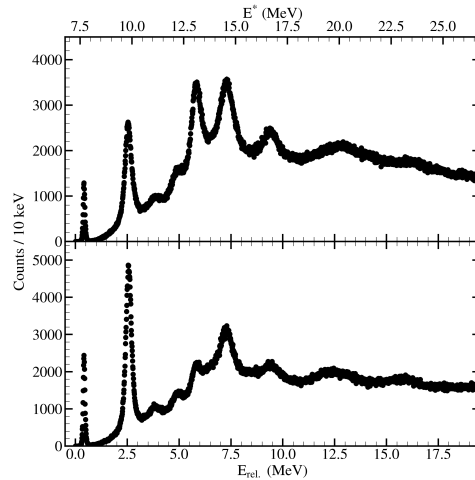
- 68 Si-CsI(Tl) telescopes
 - Excellent forward coverage ($1.7^\circ - \sim 40^\circ$)
 - Lab energy (Thick CsI stops particles)
 - Isotopic particle identification ($E-\Delta E$ technique)
 - Lab angle (Dual-Axis Duo-Lateral detectors)



Two Active Studies

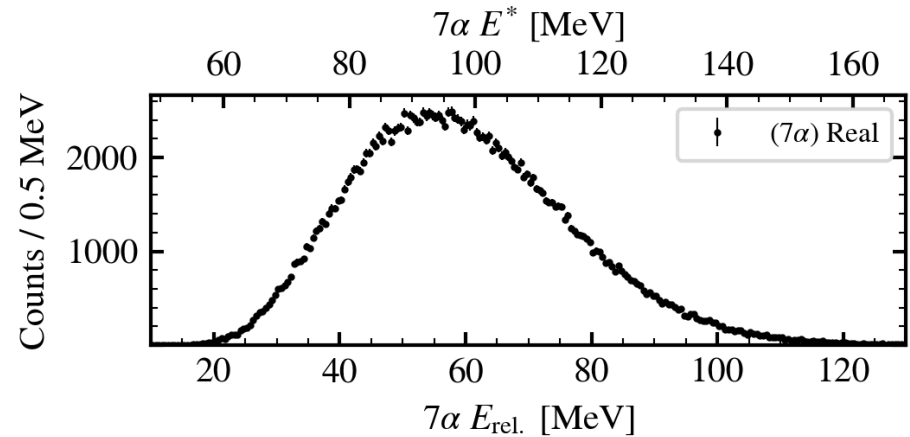
Some states of $^{12}\text{C}^*$ are observed to decay into 3α .

How does the population of these states change for two similar reactions?



Predictions for exotic α -clustered structure in $^{28}\text{Si}^*$ have been predicted.

Is there any evidence of $^{28}\text{Si}^* \rightarrow 7\alpha$ resonant decays?



Both studies (and many more) need a detailed characterization of the nonresonant background.

Characterizing the Nonresonant Background

A Case Study on d - d - d

- Any signals should be **rare to nonexistent**

- $Q(^6\text{Li} \rightarrow 3d) = -25.3 \text{ MeV}$

- $Q(\alpha \rightarrow 2d) = -23.8 \text{ MeV}$

- $Q(d \rightarrow pn) = -2.2 \text{ MeV}$

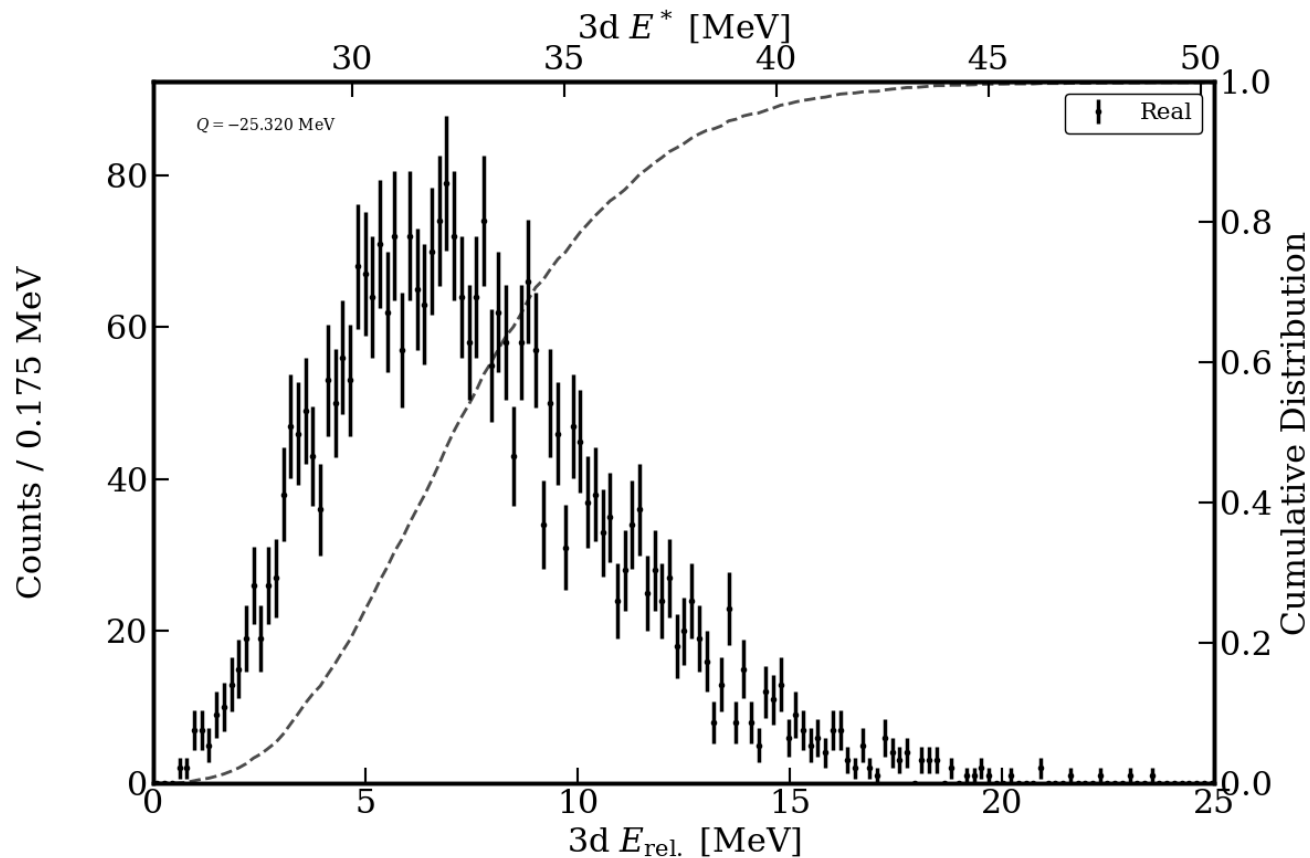
$^6\text{Li}(E^* > 25.32 \text{ MeV})$

$\rightarrow \alpha^*(E^* > 23.8 \text{ MeV})$

$+ d(E^* < 2.2 \text{ MeV})$

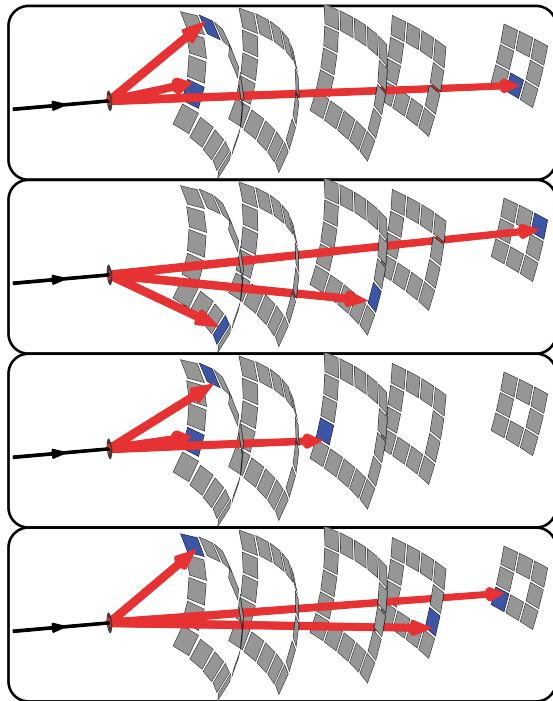
$\rightarrow 3d(E^* < 2.2 \text{ MeV})$

Total = Background + Signal



Creating a Mixed Event

Real Events

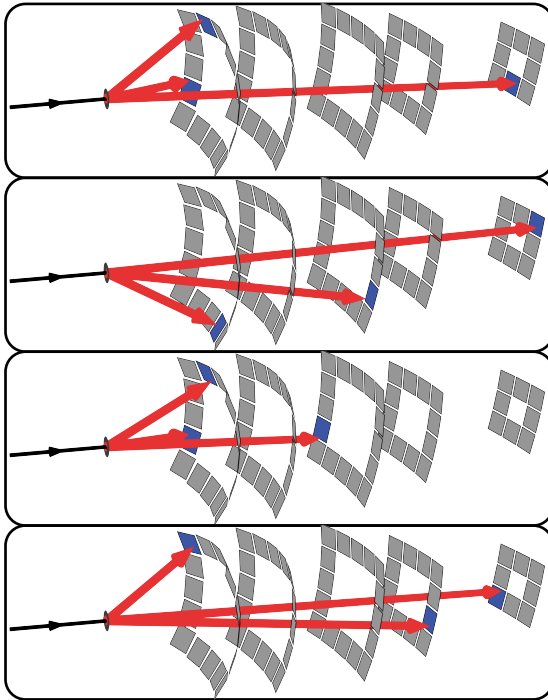


⋮

- Start w/ events containing all particles

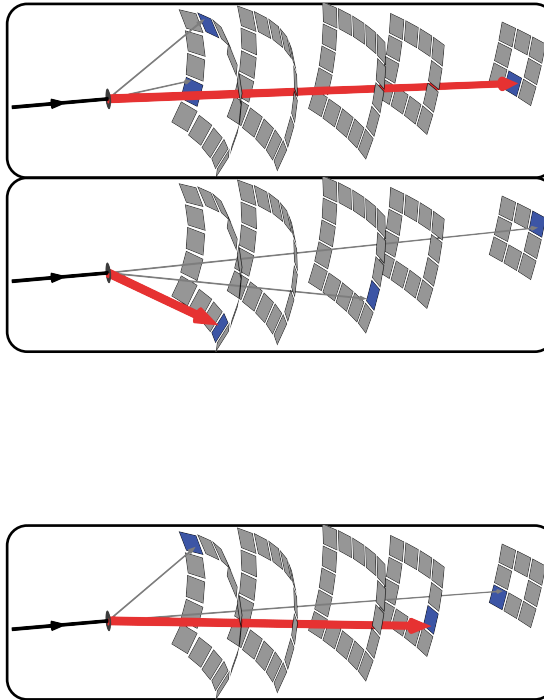
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Real Events



⋮

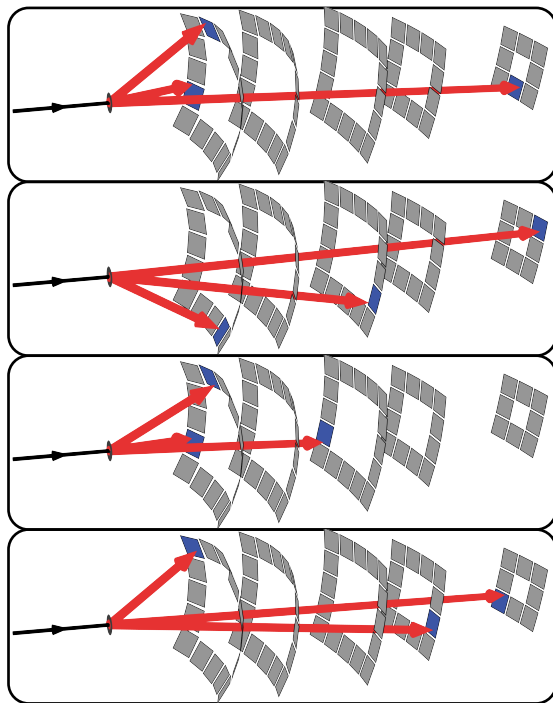
Selected Events



- Start w/ events containing all particles
- Select particles from separate events

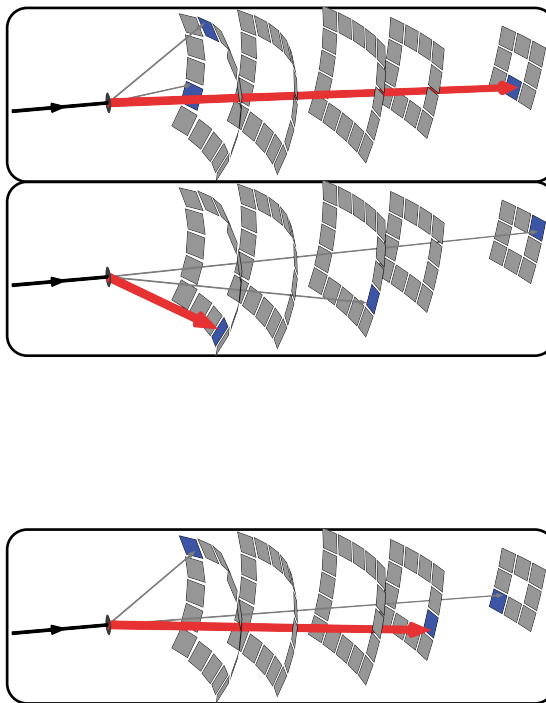
Creating a Mixed Event

Real Events

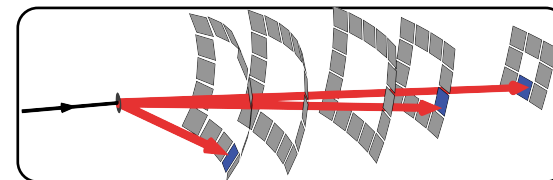


⋮

Selected Events



Mixed Events



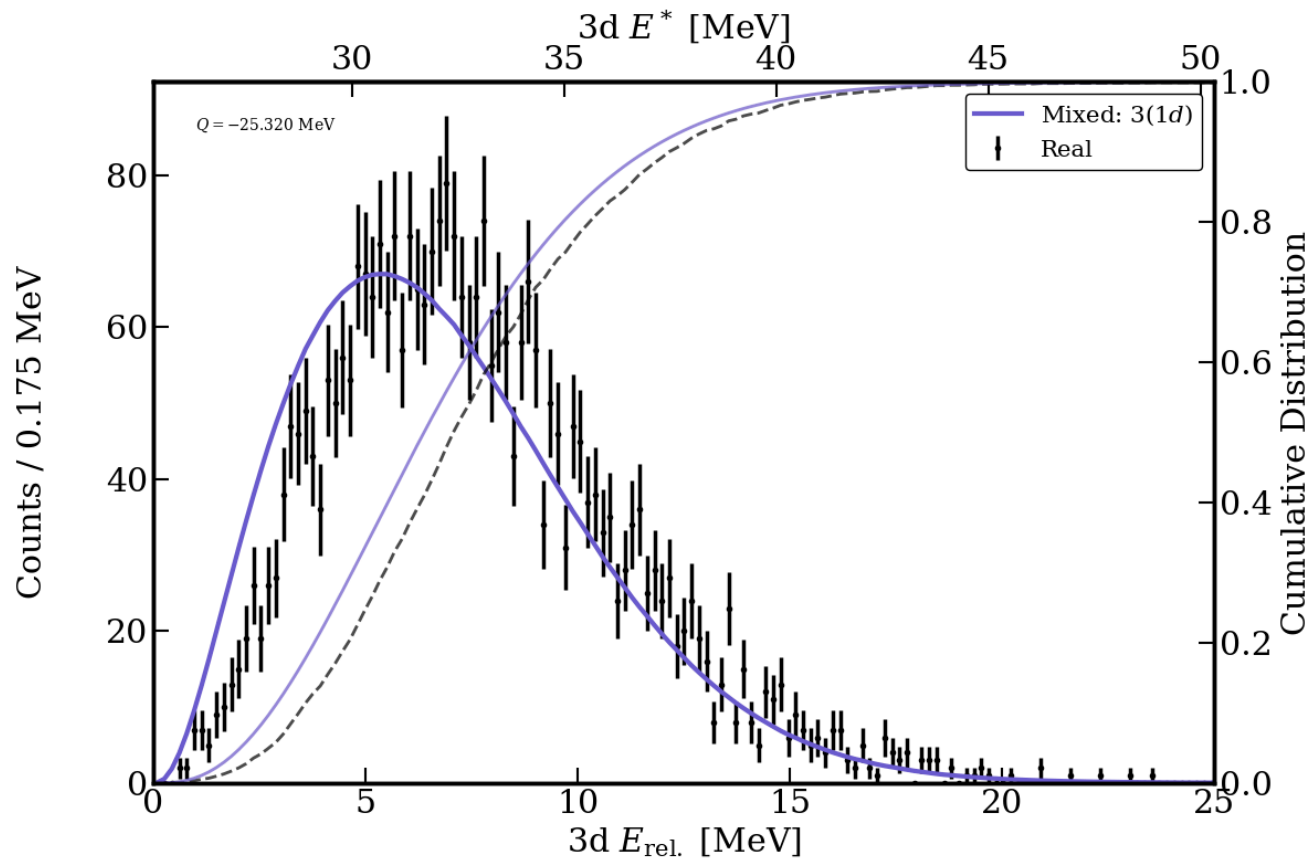
⋮

- Start w/ events containing all particles
- Select particles from separate events
- Construct new event

One way we know mixed events are imperfect (for this purpose)

Mixed $\neq$ Total

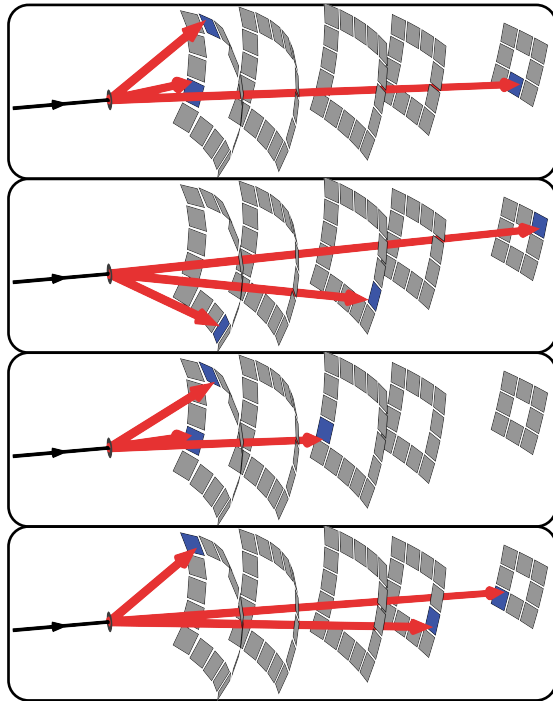
- Mixing events removes particle-particle correlations, including:
 - 3-particle resonances (good)
 - Coulomb repulsion (bad)
 - 2-particle resonances (bad)
- Need a way to **remove 3-particle** correlations while **preserving 2-particle** correlations



Folding in Lower-Order Correlations

Creating a *Partially Mixed Event*

Real Events

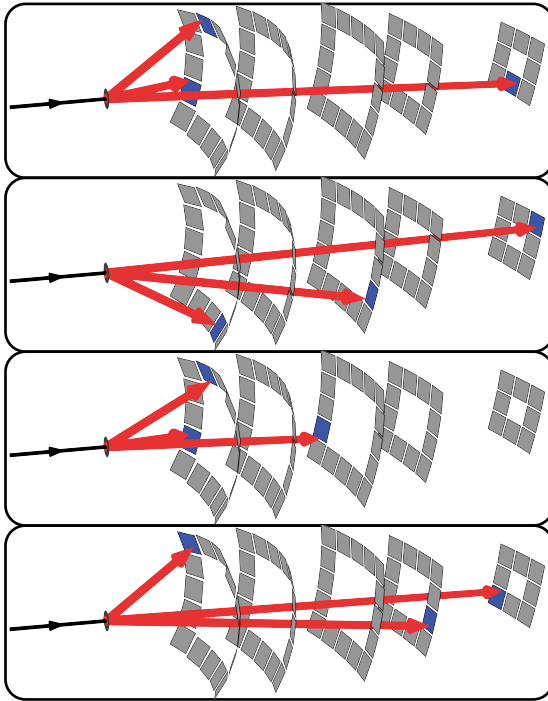


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- Start w/ events containing all particles

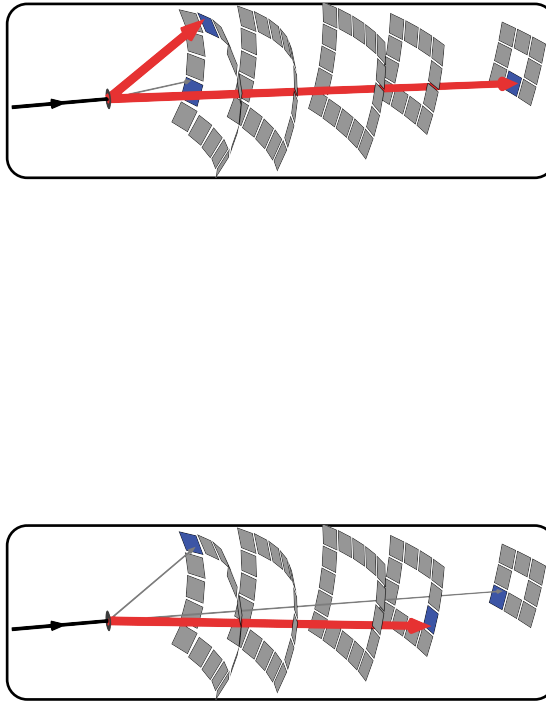
Creating a *Partially Mixed Event*

Real Events



⋮

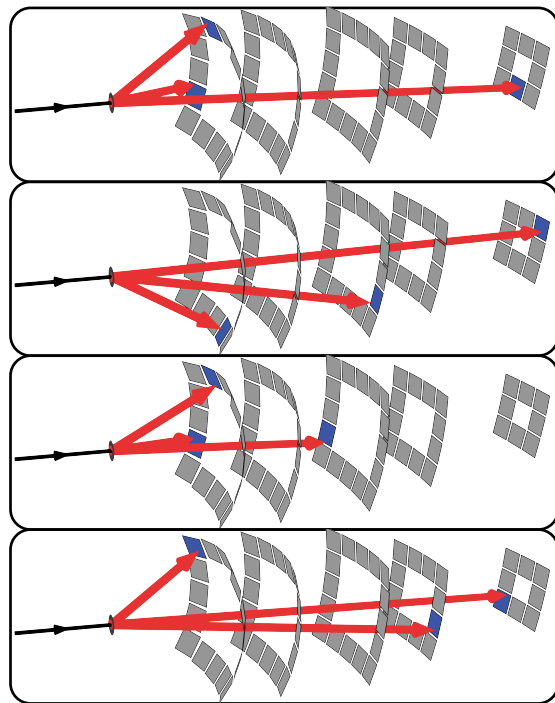
Selected Events



- Start w/ events containing all particles
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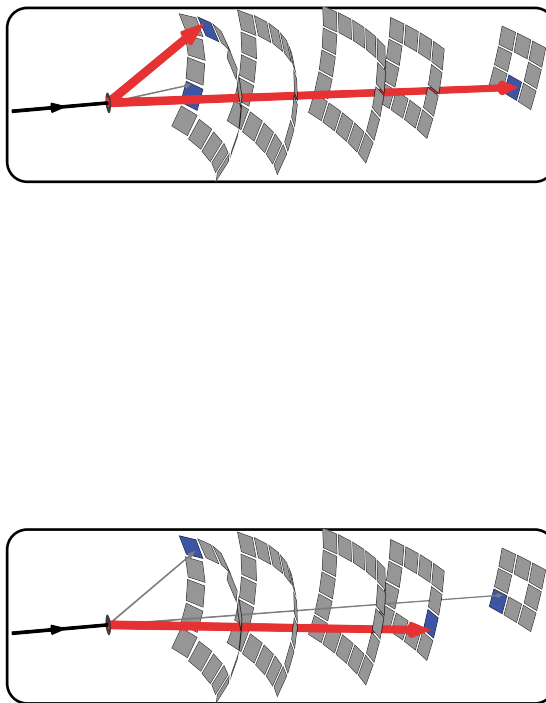
Creating a *Partially Mixed Event*

Real Events

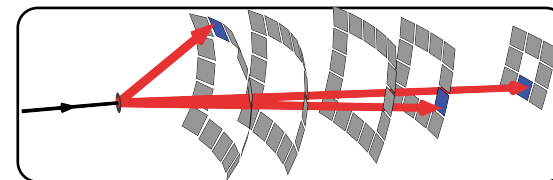


⋮

Selected Events



Partially Mixed Events

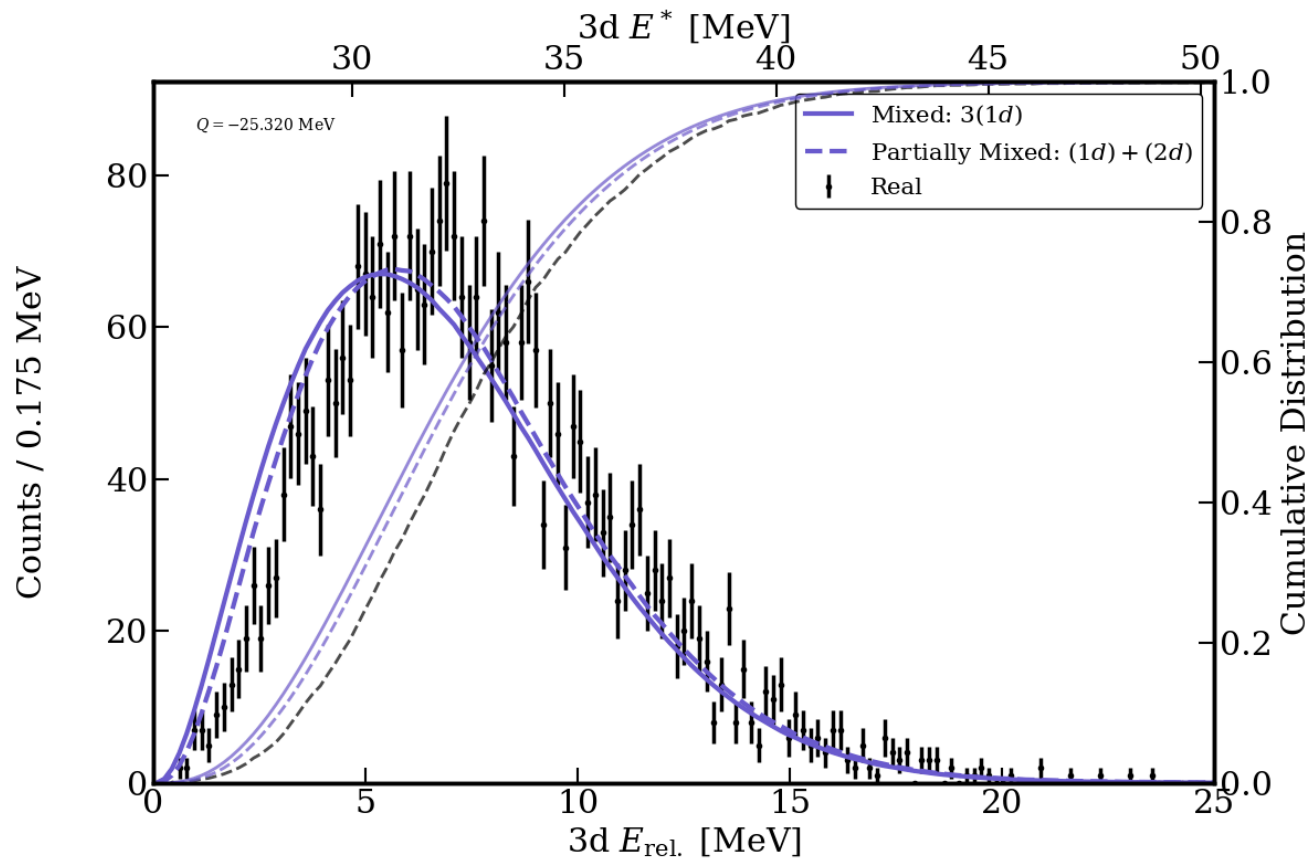


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- Start w/ events containing all particles
- Select particles from separate events
- Construct new event

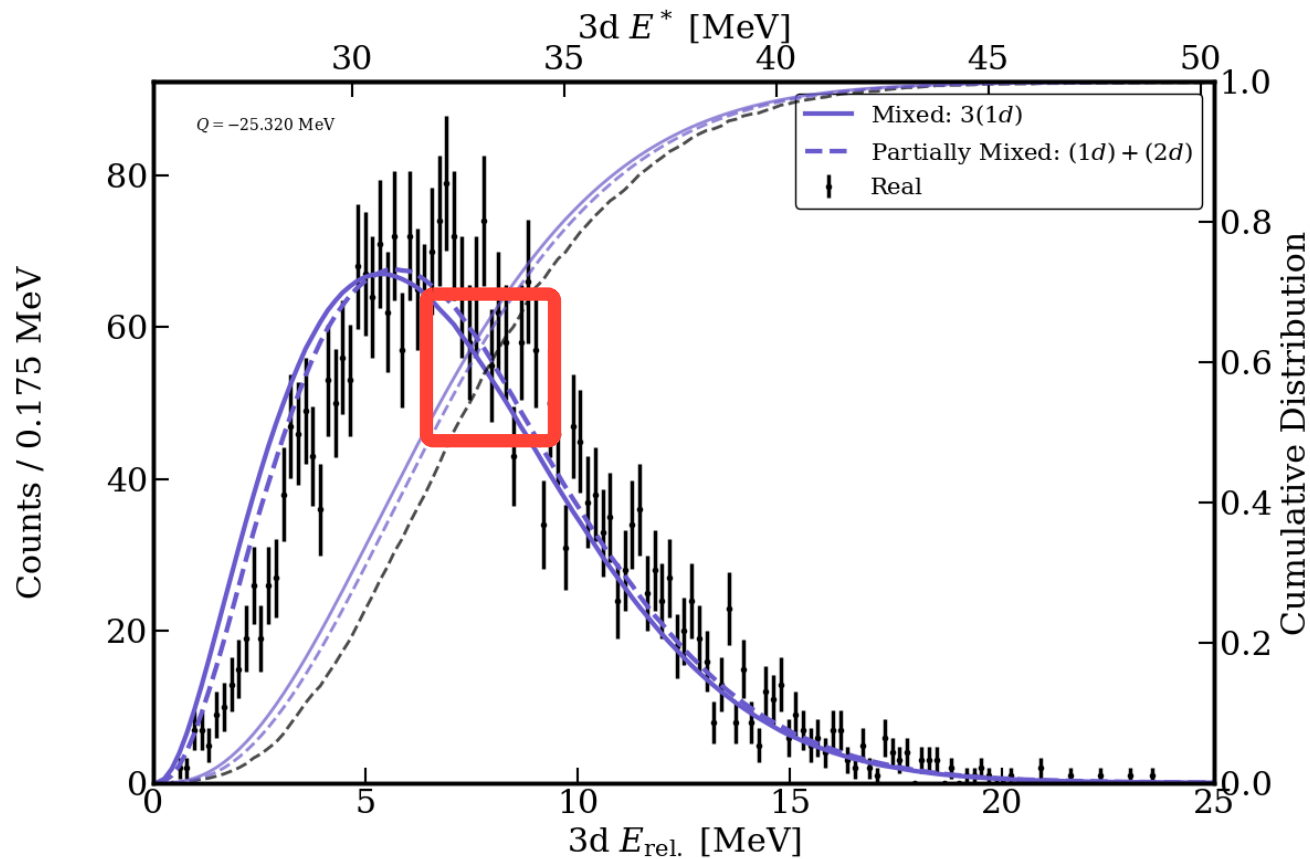
Incorporating one of the d - d correlations

- Partial mixing (PM) provides a minor improvement over full mixing (FM).
- PM is $\sim \frac{1}{3}$ of the way between FM and Real in the inverse CDF



Incorporating one of the d - d correlations

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Measure the impact of one $d-d$ correlation

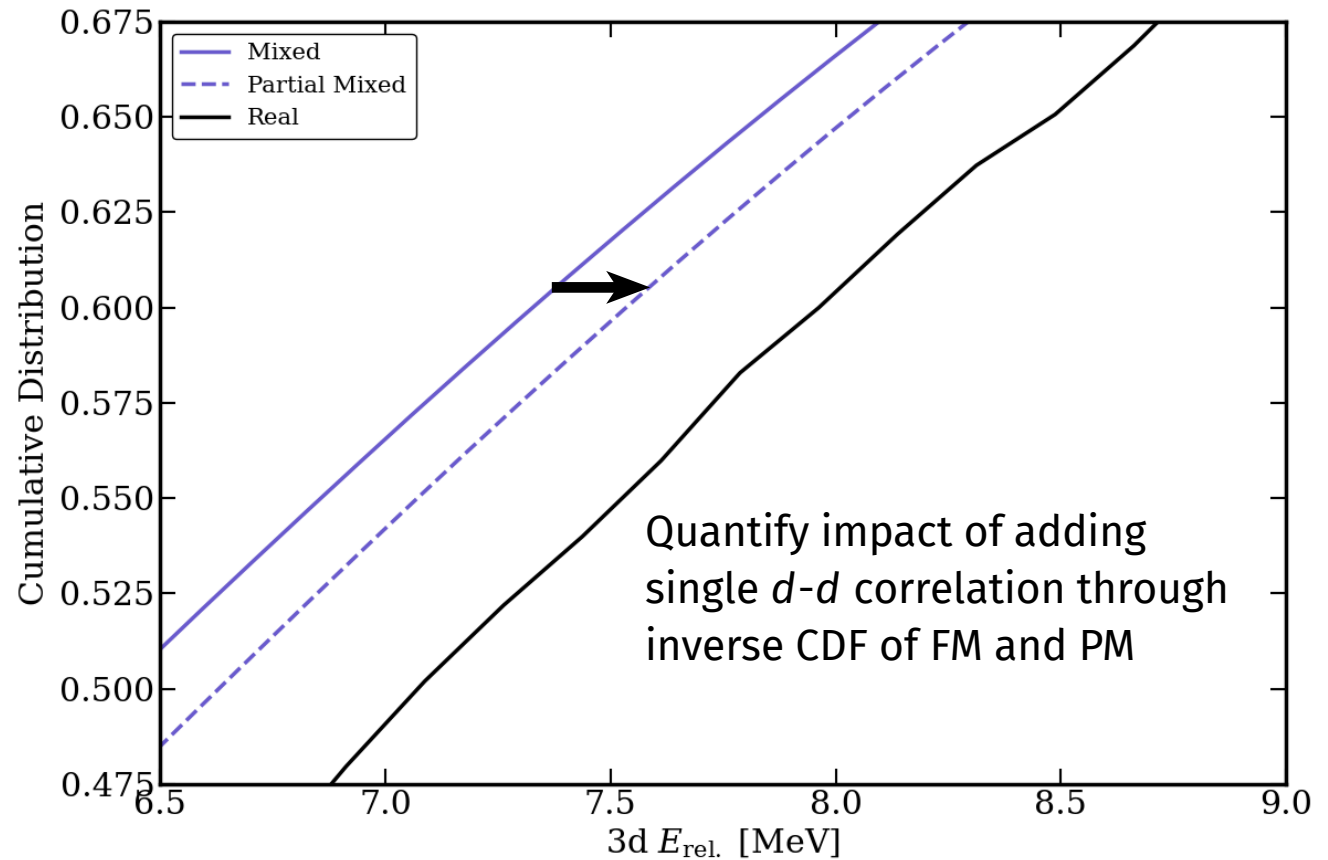
Number of $d-d$ correlations

Real: $\binom{3}{2} = 3$

Partially Mixed: $\binom{2}{2} + \binom{1}{2} = 1$

Mixed: $3\binom{1}{2} = 0$

- Quantile mapping: Horizontal shifts in the CDF
 - Frequently used in climate change studies



Extrapolate the impact to three $d-d$ correlations

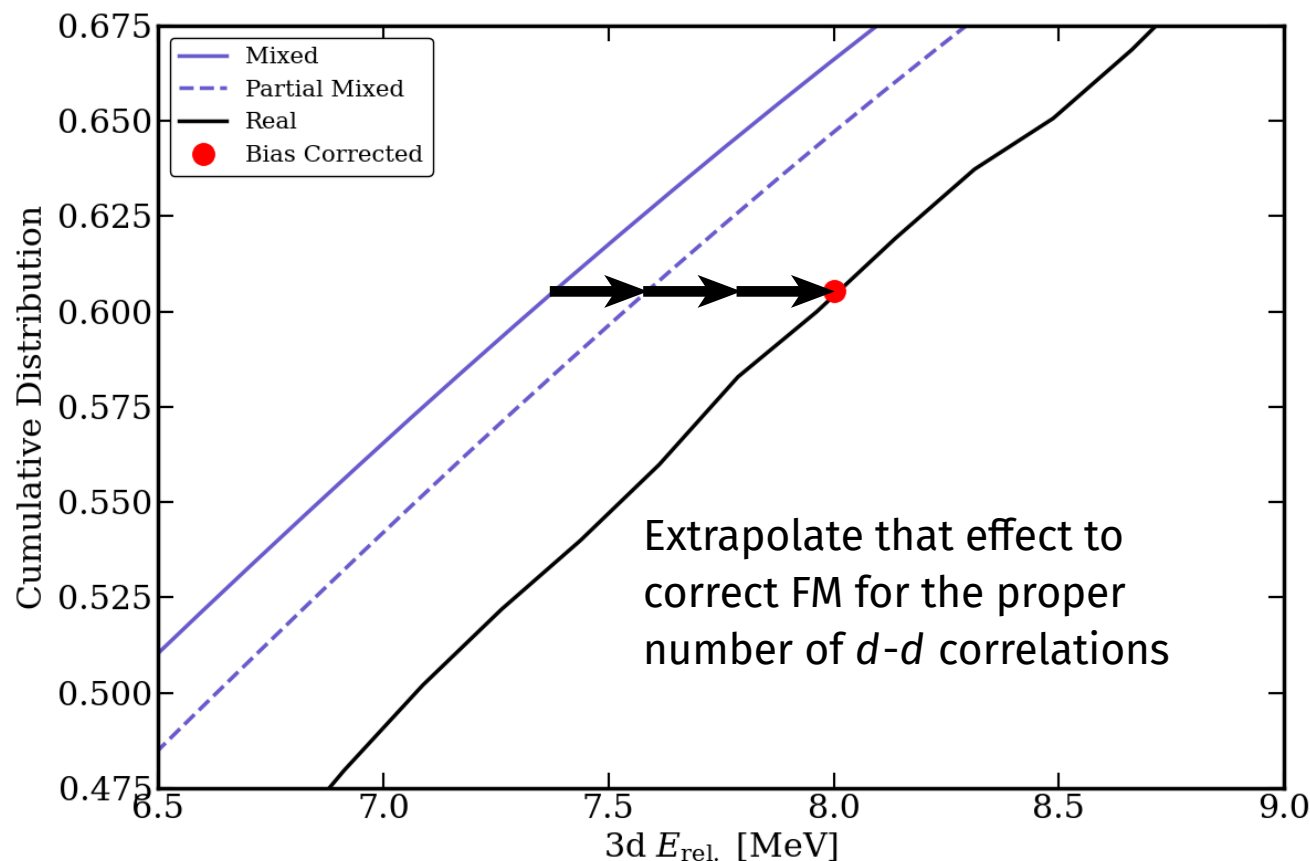
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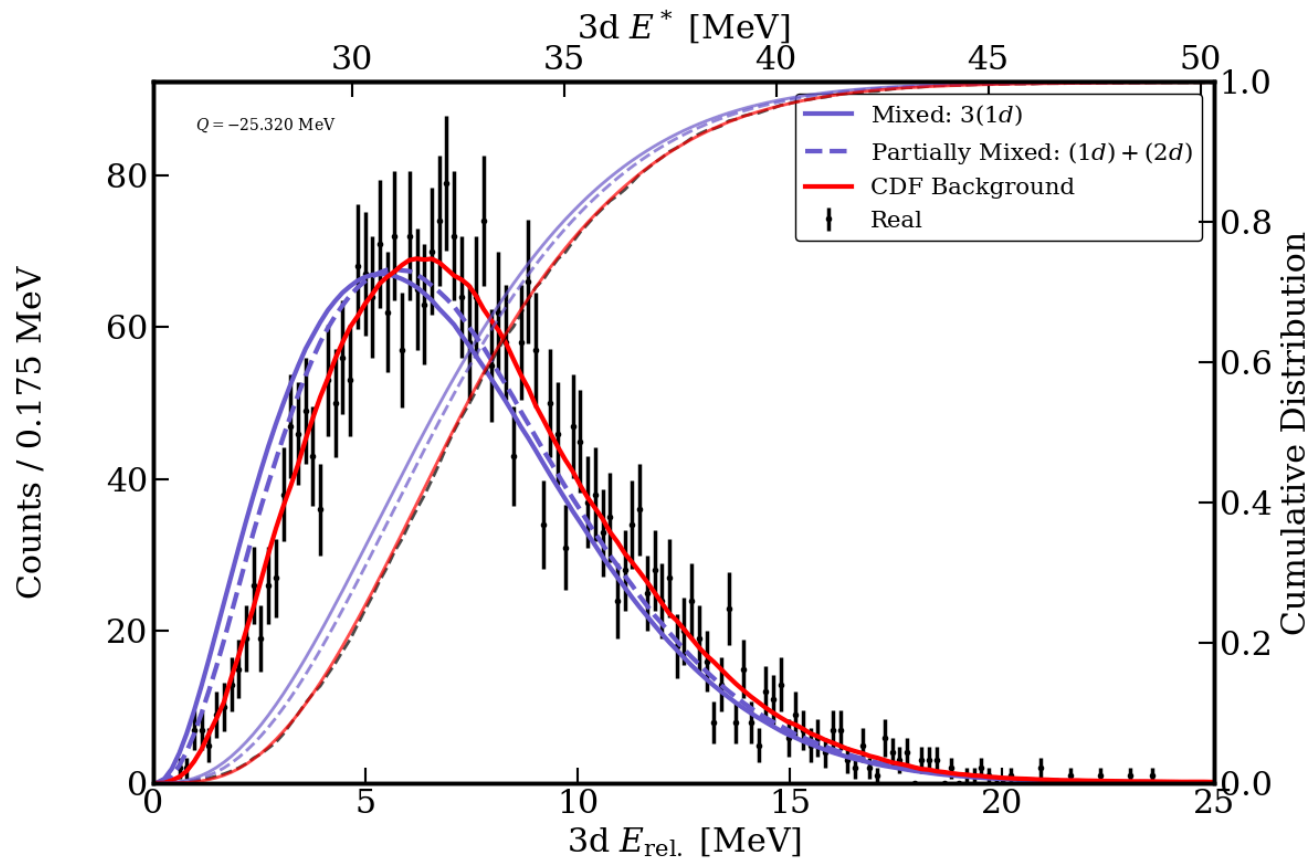
- Quantile mapping: Horizontal shifts in the CDF
 - Frequently used in climate change studies



Derive new background through CDF

Estimation of the Nonresonant Background

1. Measure the systematic impact of introducing a single 2-particle correlation, via the transformation from FM to PM
2. Propagate that systematic change an additional 2 times to account for all three 2-particle correlations



Derive new background through CDF

Estimation of the Nonresonant Background

$B_{ddd}(E)$ Background estimate for ddd

$M_{\emptyset}(E)$ Fully Mixed Distribution (No Preservation)

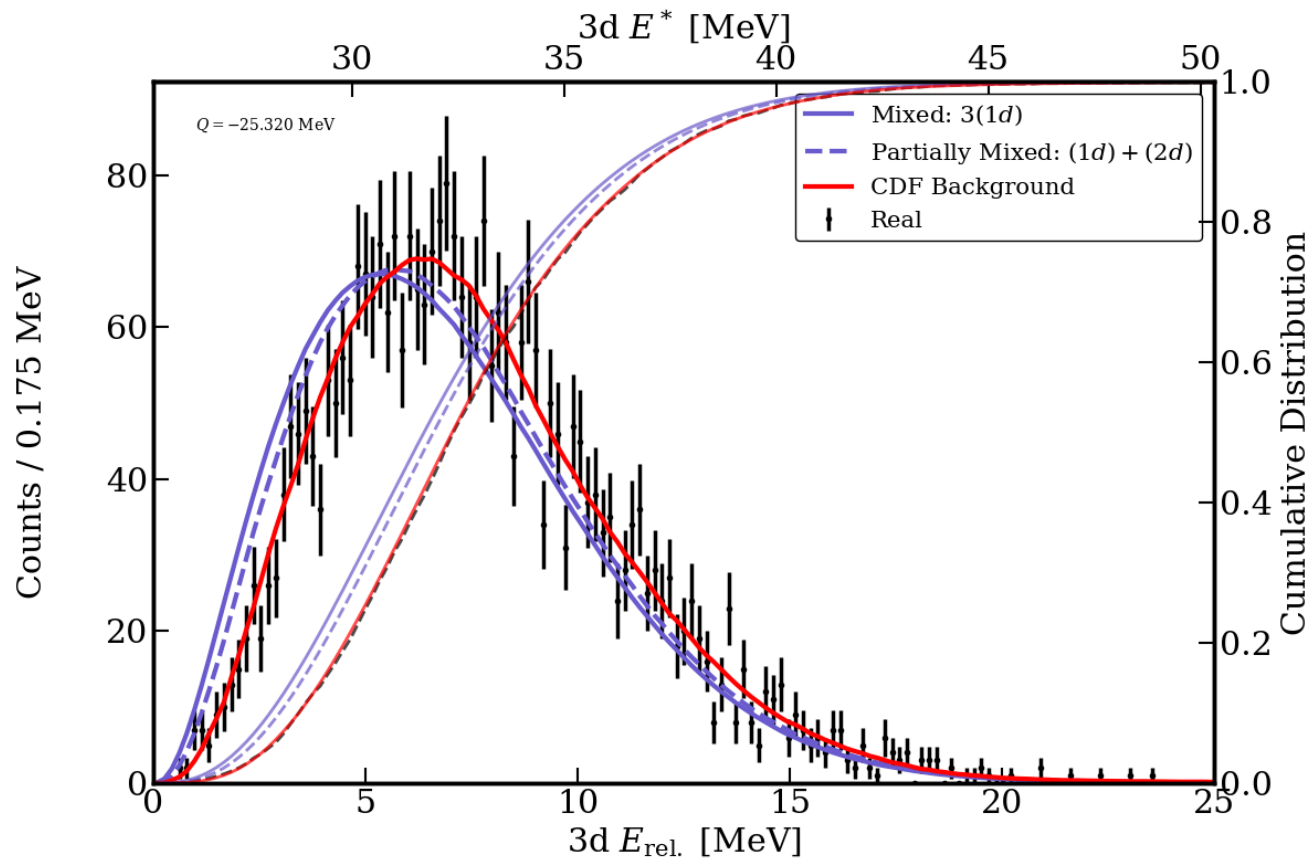
$M_{dd}(E)$ Partially Mixed Distribution (dd Preserved)

$\mathcal{B}_{ddd}^{-1}(q)$ Inverse CDF of $B_{ddd}(E)$

$\mathcal{M}_{\emptyset}^{-1}(q)$ Inverse CDF of $M_{\emptyset}(E)$

$\mathcal{M}_{dd}^{-1}(q)$ Inverse CDF of $M_{dd}(E)$

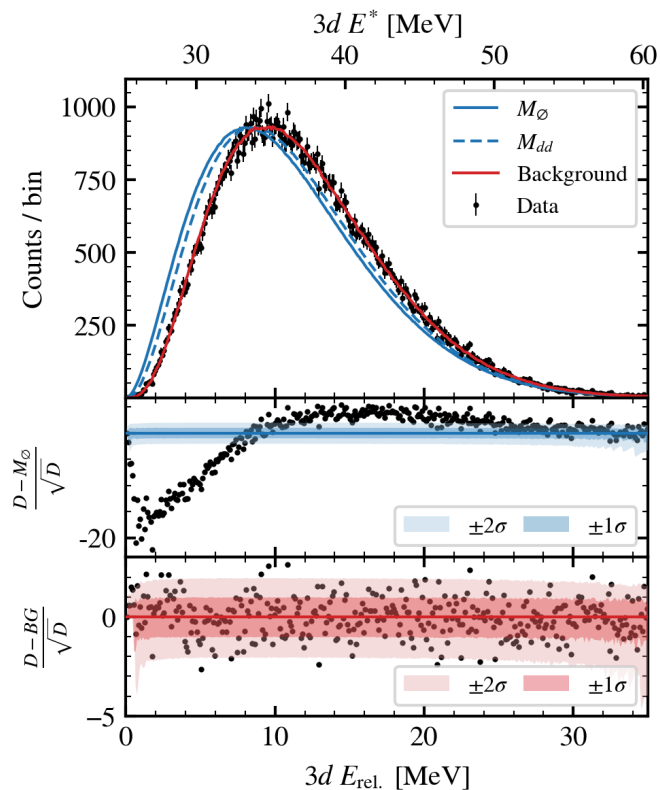
$$\mathcal{B}_{ddd}^{-1}(q) = \mathcal{M}_{\emptyset}^{-1}(q) + 3[\mathcal{M}_{dd}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)]$$



Three 3d Correlations

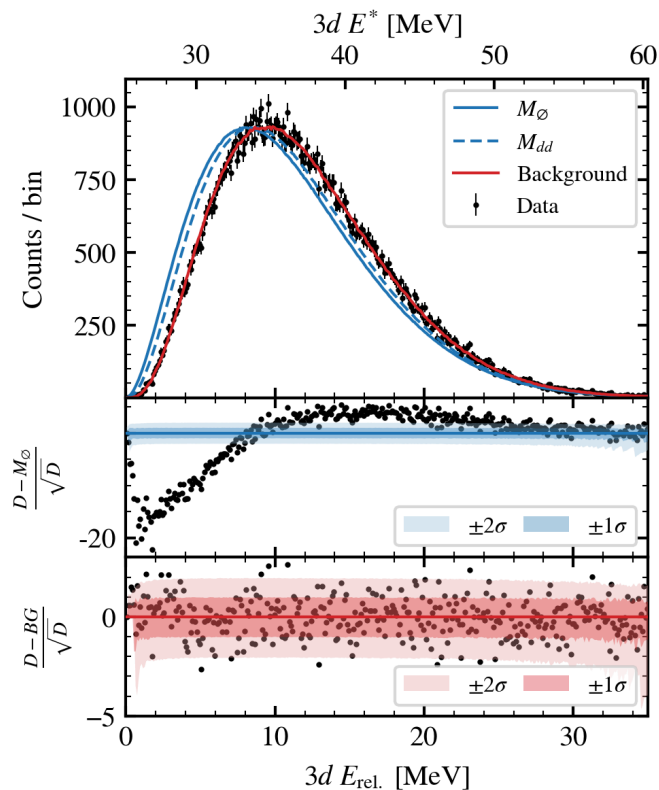
FAUST

$^{16}\text{O} + ^{12}\text{C}$ @ 28.3 MeV



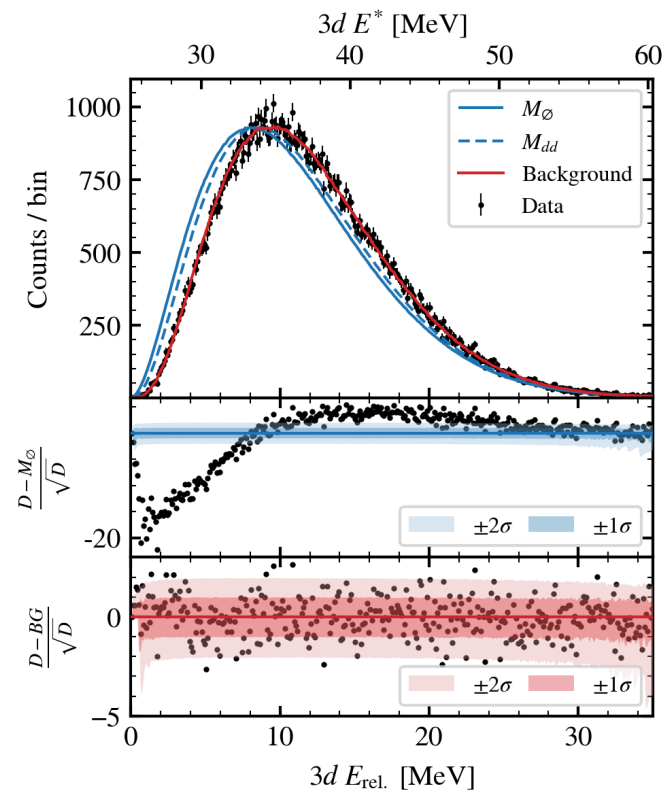
FAUST

$^{14}\text{N} + ^{10}\text{B}$ @ 28.3 MeV



NIMROD

$^{28}\text{Si} + ^{12}\text{C}$ @ 35 MeV



Generalization of the Method (So Far) - TODO Cartoons

3 Identical Particles

$$\mathcal{B}_{XXX}^{-1}(q) = \mathcal{M}_{\emptyset}^{-1}(q) + 3[\mathcal{M}_{XX}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)]$$

3 Non-identical Particles

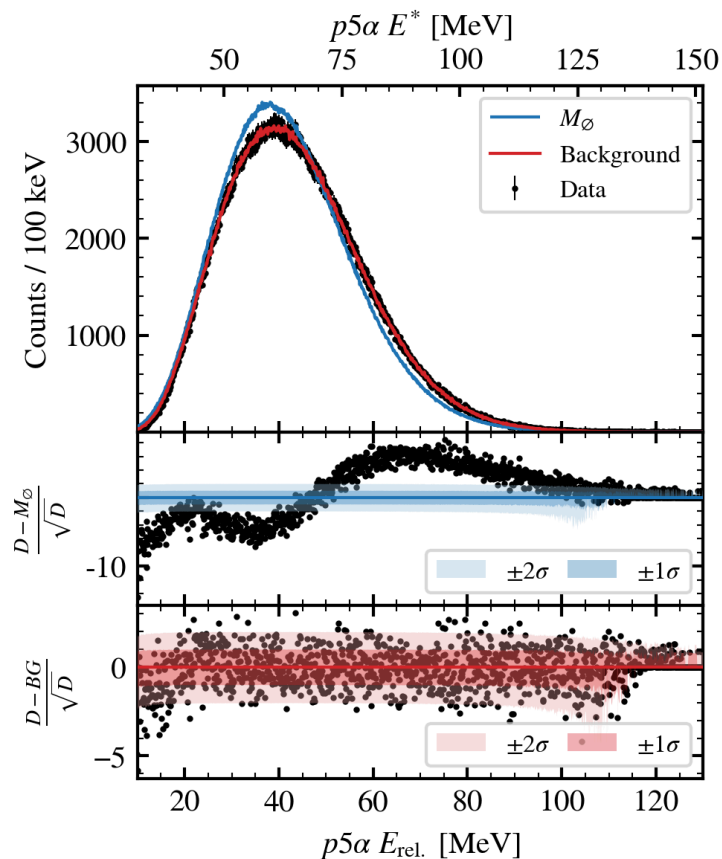
$$\begin{aligned}\mathcal{B}_{XYZ}^{-1}(q) = & \mathcal{M}_{\emptyset}^{-1}(q) + [\mathcal{M}_{XY}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)] \\ & + [\mathcal{M}_{YZ}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)] \\ & + [\mathcal{M}_{XZ}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)]\end{aligned}$$

- P is a multiset of particles (e.g., $P = [d, d, d]$)
- $M_{P \setminus \{i\}}(E)$ is the energy distribution for mixed events where particle i was swapped.

3 or More Generic Particles

$$\mathcal{B}_P^{-1}(q) = \mathcal{M}_{\emptyset}^{-1}(q) + \frac{1}{|P| - 2} \sum_{i \in P} [\mathcal{M}_{P \setminus \{i\}}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)]$$

Example with $p5\alpha$ - todo cartoons... probably



$$\mathcal{B}_P^{-1}(q) = \mathcal{M}_{\emptyset}^{-1}(q) + \frac{1}{|P| - 2} \sum_{i \in P} [\mathcal{M}_{P \setminus \{i\}}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)]$$

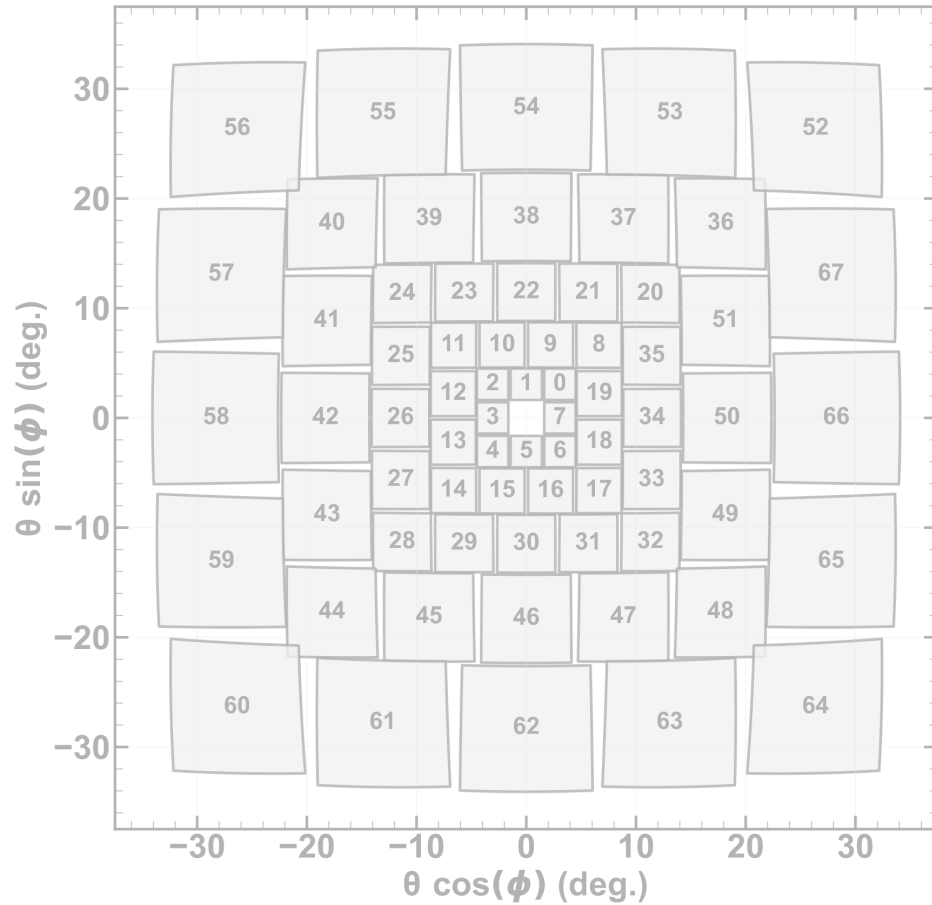
- Particle set:
 - $P = [p, \alpha, \alpha, \alpha, \alpha, \alpha]$
 - $|P| - 2 = 4$

- Background:

$$\begin{aligned} \mathcal{B}_{p5\alpha}^{-1}(q) &= \mathcal{M}_{\emptyset}^{-1}(q) \\ &\quad + \frac{1}{4} [\mathcal{M}_{5\alpha}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)] \\ &\quad + \frac{5}{4} [\mathcal{M}_{p4\alpha}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)] \end{aligned}$$

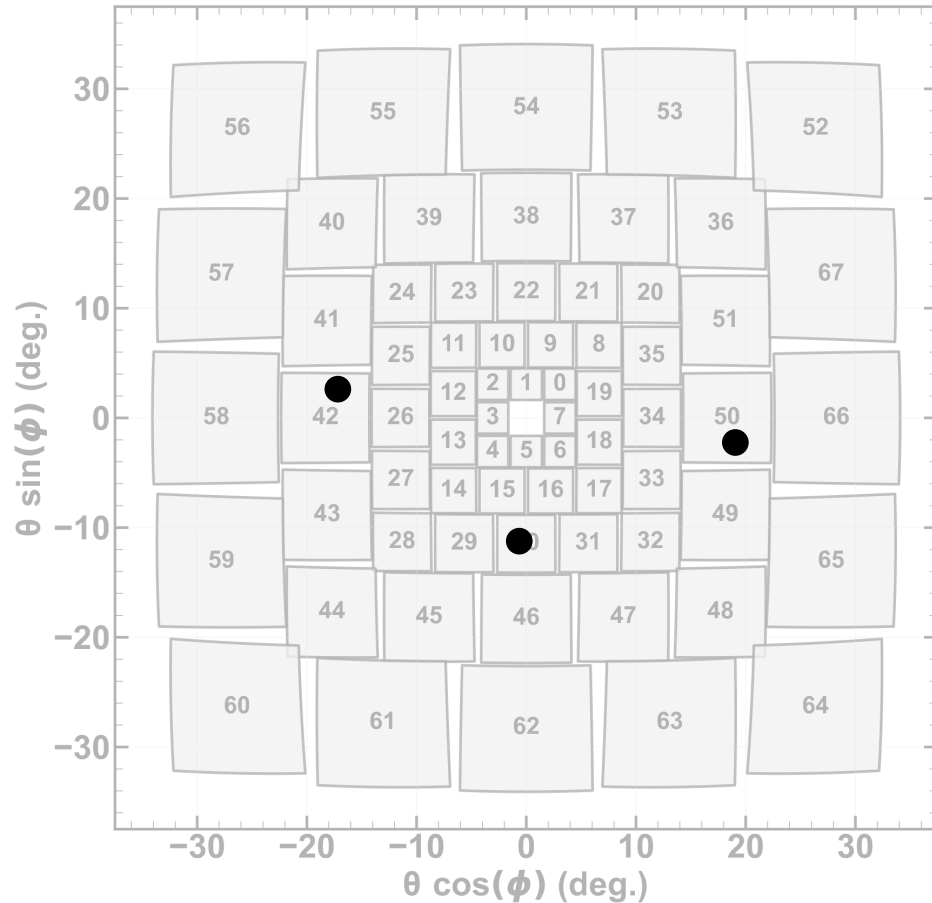
- Standard mixing fails to describe the distribution
- Current work gets **much** closer, but still slightly off at low energy

Another Problem with Traditional Mixing



- Consider 3 identical particles again
- Standard Partial Mixing Procedure

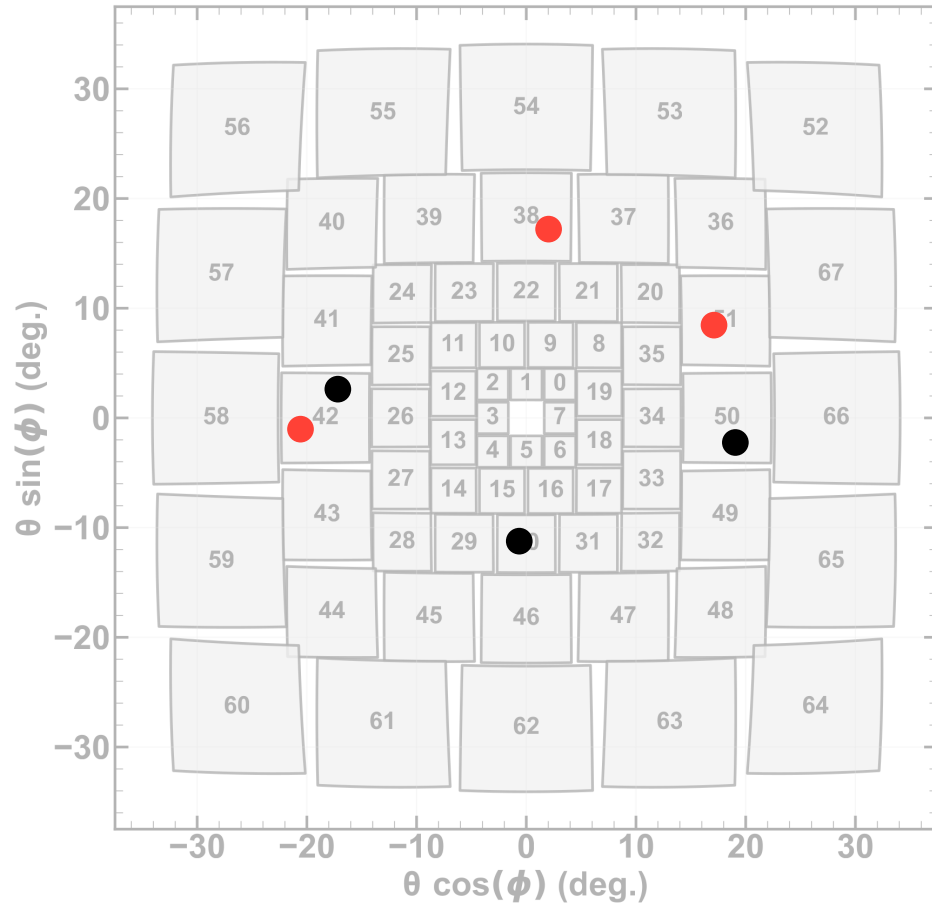
Another Problem with Traditional Mixing



Consider 3 identical particles again

- Standard Partial Mixing Procedure
 - Select one **event**

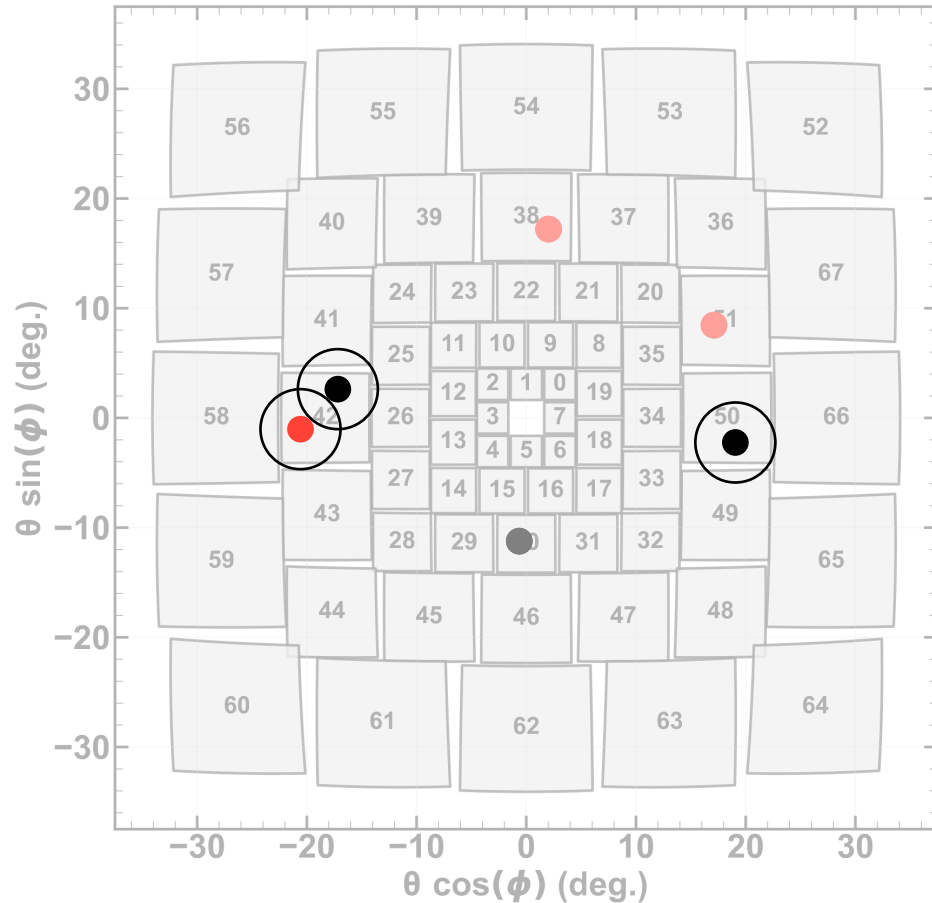
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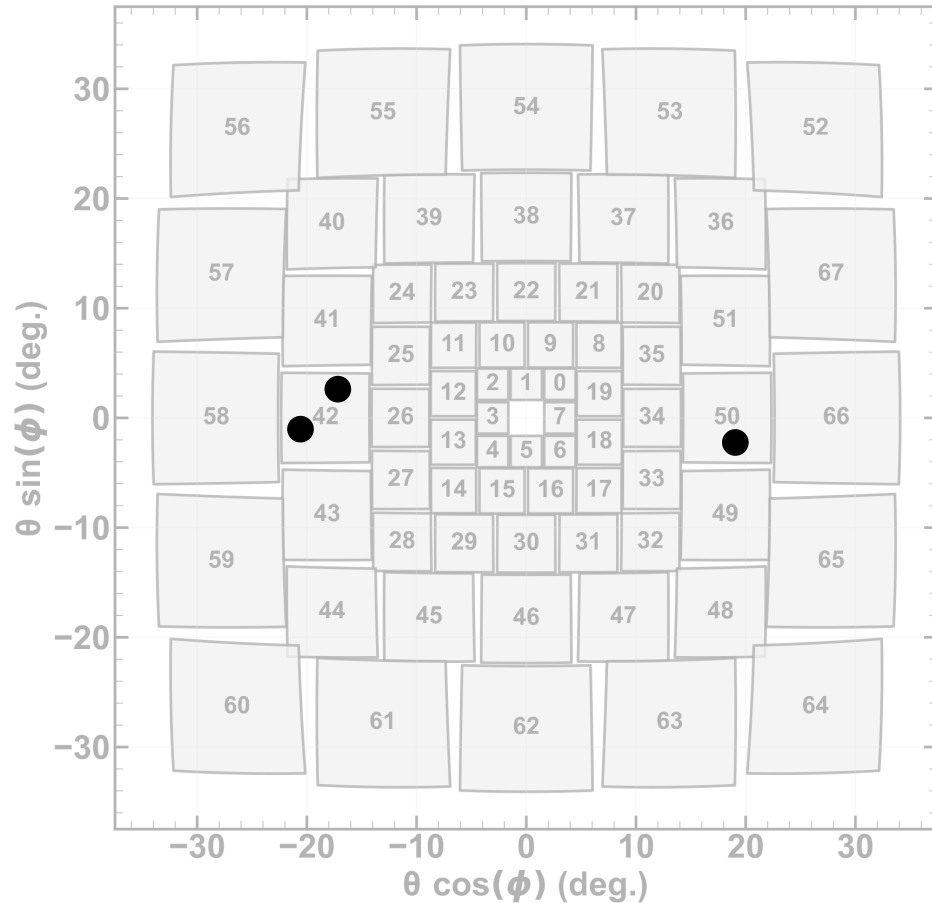
Another Problem with Traditional Mixing



Consider 3 identical particles again

- Standard Partial Mixing Procedure
 - Select one **event**
 - Select another **event**
 - Choose all but one particle from one **event** and the last particle from **the other event**

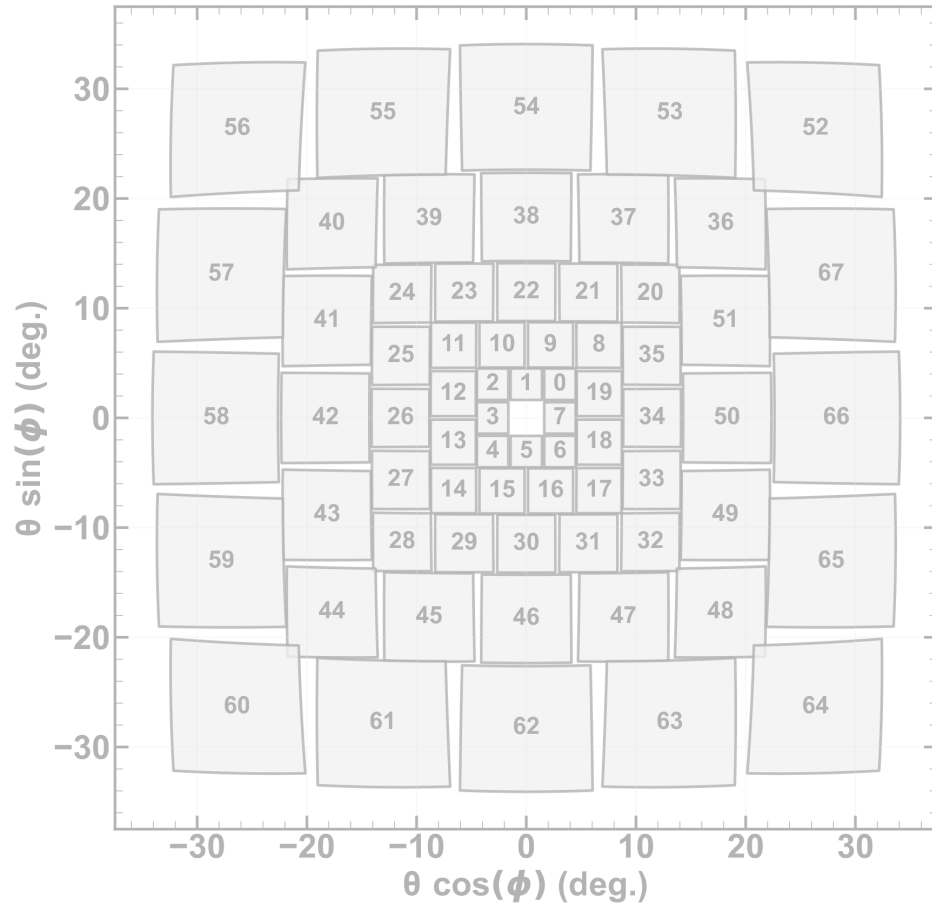
Another Problem with Traditional Mixing



Consider 3 identical particles again

- Standard Partial Mixing Procedure
 - Select one **event**
 - Select another **event**
 - Choose all but one particle from one **event** and the last particle from **the other event**
 - New event has a double hit!

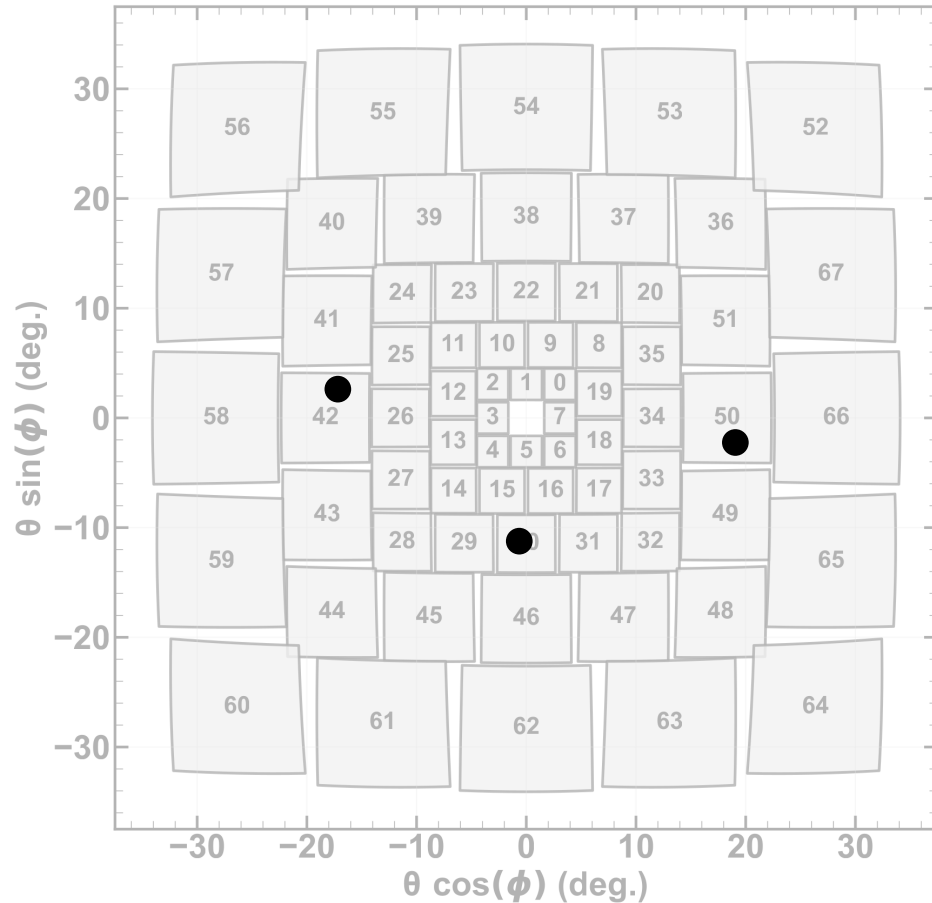
Hit Pattern Preservation (Alternate Mixing Technique)



Consider 3 identical particles again

- Hit Pattern Preserving Partial Mixing Procedure

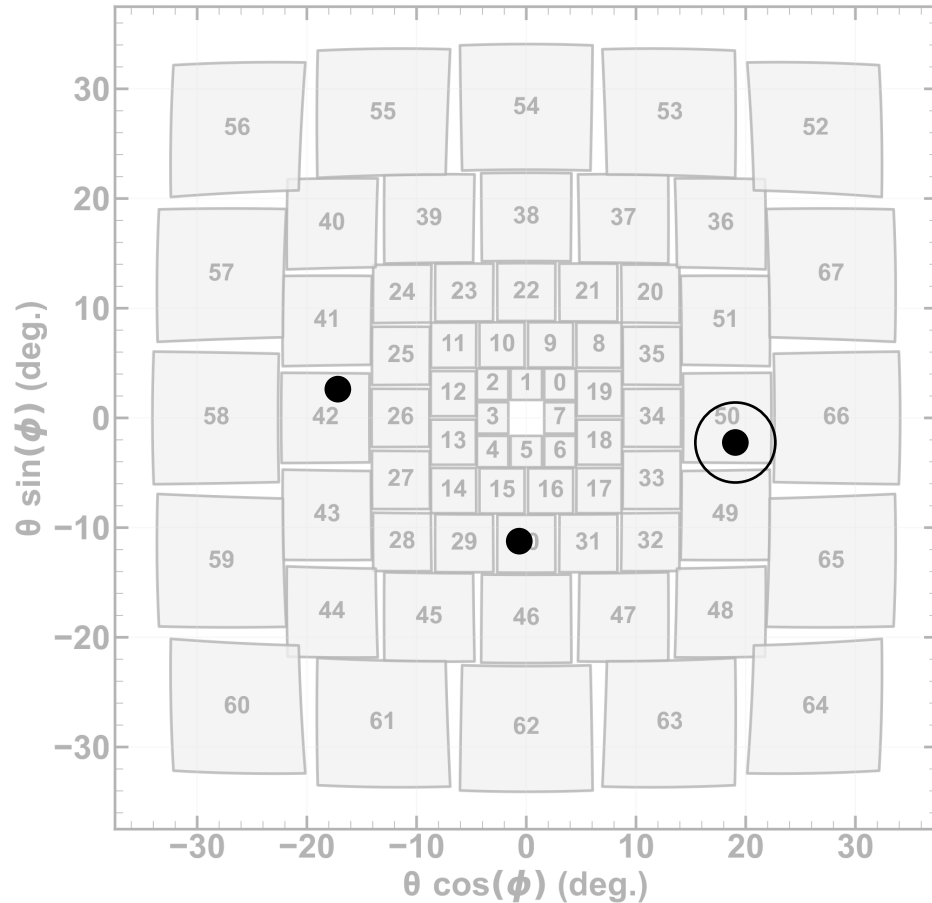
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Consider 3 identical particles again

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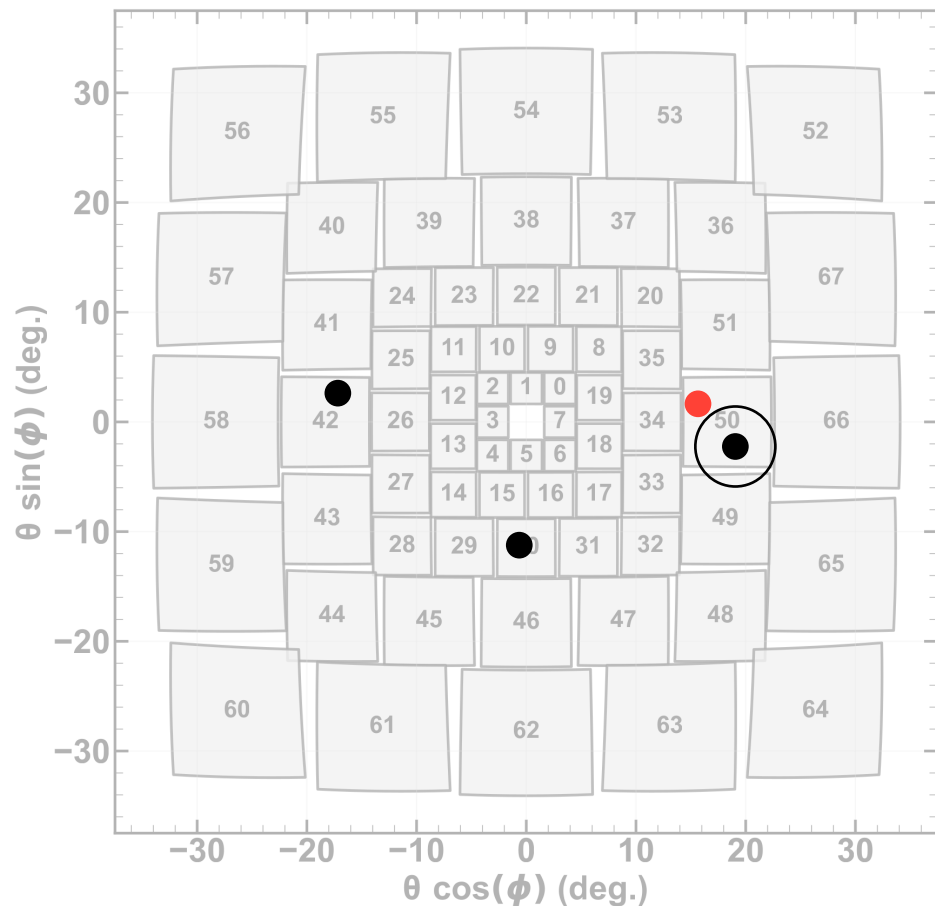
Hit Pattern Preservation (Alternate Mixing Technique)



Consider 3 identical particles again

- Hit Pattern Preserving Partial Mixing Procedure
 - Select one **event**
 - Identify particle that is *not* to be preserved

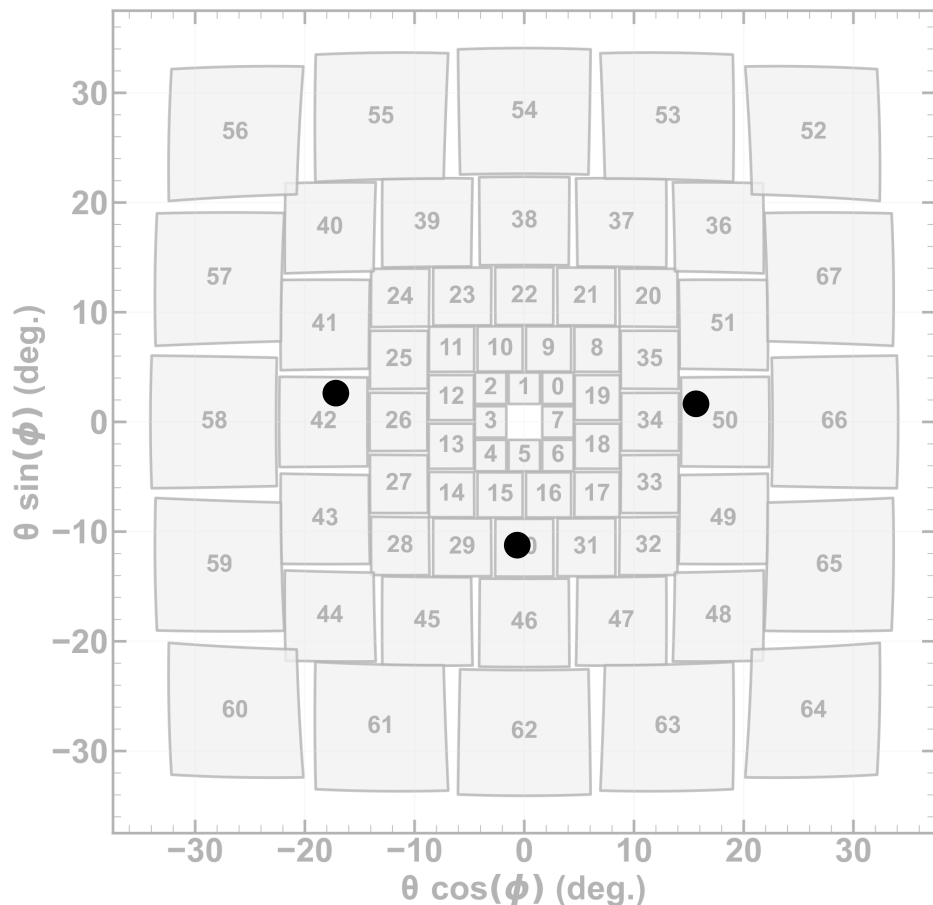
Hit Pattern Preservation (Alternate Mixing Technique)



Consider 3 identical particles again

- Hit Pattern Preserving Partial Mixing Procedure
 - Select one **event**
 - Identify particle that is *not* to be preserved
 - Find a particle of same type that was measured in another **event** by the *same detector*

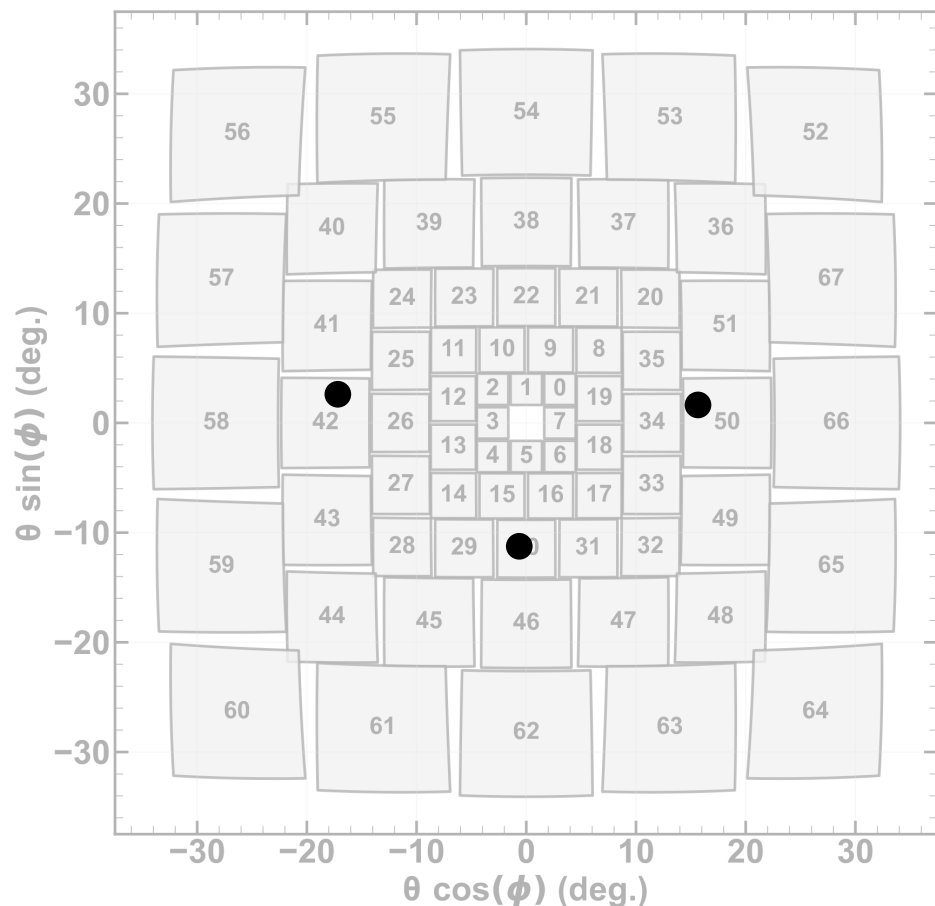
Hit Pattern Preservation (Alternate Mixing Technique)



Consider 3 identical particles again

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 - ▶ Select one **event**
 - ▶ Identify particle that is *not* to be preserved
 - ▶ Find a particle of same type that was measured in another **event** by the *same detector*
 - ▶ New event has no double hits!
 - Has exactly same hit pattern as real
 - ▶ For Full Mixing:
 - Swap each particle individually from all different events

Hit Pattern Preservation (Alternate Mixing Technique)

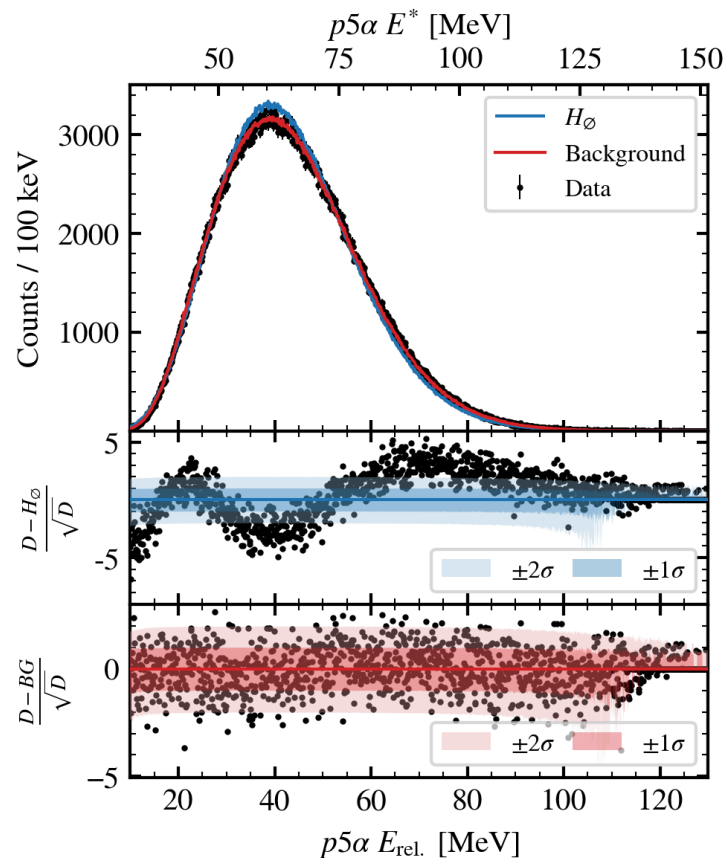


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 - For Full Mixing:
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$$\mathcal{B}_P^{-1}(q) = \mathcal{H}_{\emptyset}^{-1}(q) + \sum_{i \in P} w_i [\mathcal{H}_{P \setminus \{i\}}^{-1}(q) - \mathcal{H}_{\emptyset}^{-1}(q)]$$

Example with $p5\alpha$ - todo cartoons... probably



$$\mathcal{B}_P^{-1}(q) = \mathcal{H}_\emptyset^{-1}(q) + \sum_{i \in P} w_i [\mathcal{H}_{P \setminus \{i\}}^{-1}(q) - \mathcal{H}_\emptyset^{-1}(q)]$$

- Particle set:

- $P = [p, \alpha, \alpha, \alpha, \alpha]$
- $|P| - 2 = 4$

- Background:

$$\begin{aligned} \mathcal{B}_{p5\alpha}^{-1}(q) = & \mathcal{M}_\emptyset^{-1}(q) \\ & + 0.248 [\mathcal{M}_{5\alpha}^{-1}(q) - \mathcal{M}_\emptyset^{-1}(q)] \\ & + 1.045 [\mathcal{M}_{p4\alpha}^{-1}(q) - \mathcal{M}_\emptyset^{-1}(q)] \end{aligned}$$

- $H_\emptyset$ is not enough to describe the background.
- $H_\emptyset$, $H_{p4\alpha}$, and $H_{5\alpha}$ in tandem are enough to describe the distribution very well.

Systems with Resonances

pta with hpp no-signal

- doesn't work as a fit or a background
- Explain traditional fit

One of the two-d plots illustrating signal suppression

All of the 2d plots

Vary signal description, re-estimate the background, calculate signal+background and compare to original distribution

result with pta

result with pta, paa and ppaa

Application to Toroidal Structure Search

Application to Travis